

Zexu Chen

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Research Interests

Computational materials science of structural and functional alloys; phase-field modeling of martensitic transformation and precipitation in NiTi shape memory alloys; GPU-accelerated high-performance computing; machine learning and Bayesian inference for materials property prediction and inverse alloy design.

Education

The Ohio State University, Columbus, OH

May 2024 – May 2028 (expected)

Ph.D., Materials Science and Engineering

Advisor: Prof. Yunzhi Wang

The Ohio State University, Columbus, OH

Aug 2022 – May 2024

M.S., Materials Science and Engineering

Relevant coursework: Structure and Defects in Materials; Diffusion and Interface Kinetics; Thermodynamics of Materials.

Sichuan University / The Ohio State University

Aug 2017 – May 2022

B.Sc., Materials Science and Engineering

Publications

Peer-reviewed journal articles

1. Sriram H., Feng L., **Chen Z.**, Wang Y. (2026). Soft and Hard Confinement Effects on Martensitic Transformation in NiTi Shape Memory Alloys: Insights from Phase-Field Simulations. *Shape Memory and Superelasticity*. <https://doi.org/10.1007/s40830-026-00619-3>
2. Abe O.O., Qiu Z., **Chen Z.**, Jinschek J.R., Gouma P.-I. (2021). Effect of Crystallite Size on the Low-Temperature Solid-Solid Phase Transformations in the WO₃ System. *Ceramics International* 47, 33476–33482. <https://doi.org/10.1016/j.ceramint.2021.08.254>

Manuscripts under review

- **Chen Z.**, Sriram H., Yu T., Feng L., Zhu J., Rajagopalan J., Wang Y. (2026). Programming Martensitic Transformation and Mechanical Response in NiTi Shape Memory Alloys through One-Dimensional Compositional Modulations. Under review at *Acta Materialia*. Preprint: SSRN 7302838. <https://doi.org/10.2139/ssrn.7302838>

Manuscripts in preparation

- **Chen Z.**, Wang Y. Coupling Grain Structure and Compositional Gradients to Control Variant Selection in Polycrystalline NiTi (working title). *In preparation*.
- **Chen Z.**, Hu K., Tan R., Zhang T., Wang Y. AI-Enabled Inverse Design of Superelastic Materials through Phase-Field Simulations and Machine Learning (working title). *In preparation*.

Research Experience

Graduate Researcher, Wang Group, The Ohio State University

Aug 2022 – Present

Supported by the U.S. National Science Foundation, Award No. DMR-2333551 (PI: Prof. Yunzhi Wang).

Project 1 (ongoing, May 2026 – Present): Phase-Field Modeling of Grain Boundary Concentration Modulations and Martensitic Transformation in Polycrystalline NiTi Shape Memory Alloys

Project 2 (ongoing, Nov 2025 – Present): Multiscale Modeling of Shear Band Control in Layered Metallic Glass and Crystalline Composites

- Combining phase-field modeling and shear transformation zone kinetic Monte Carlo simulations, in collaboration with Prof. Yue Fan (University of Michigan). Supported by the U.S. National Science Foundation, Award No. CMMI-2611448 (PI: Prof. Yunzhi Wang).

Project 3 (ongoing, Jul 2025 – Present): Machine Learning Accelerated Inverse Design of Compositionally Graded Shape Memory Alloys

- In collaboration with Prof. Jagannathan Rajagopalan (Arizona State University).
- Developing Gaussian Process Regression surrogates that map 1D concentration-modulation parameters to transformation temperature and stress-strain behavior, enabling accelerated inverse design of NiTi alloy compositions.
- Generating training data from GPU phase-field campaigns and benchmarking kernel choices, LASSO feature selection, and Bayesian optimization for sampling the modulation design space.

Project 4 (lead author): Programming Mechanical Response of NiTi SMA via 1D Nanoscale Concentration Modulations

- Designed and ran 3D phase-field simulations of nanoscale concentration modulations to engineer linear superelastic stress-strain response of NiTi shape memory alloys for biomedical and actuator applications.
- Investigated habit-plane and twinning-plane formation in B19' martensite using martensite crystallography and Khachaturyan microelasticity theory; correlated microstructural descriptors with macroscopic mechanical behavior.

Project 5: Taming Martensitic Transformation via Dissolution of Precipitates in NiTi SMA

- Modeled Ni_4Ti_3 precipitation and B2-to-B19' martensitic transformation via multiphase-field simulations using Cahn-Hilliard and time-dependent Ginzburg-Landau evolution equations with explicit nucleation algorithm.
- Quantified effects of dissolution-induced concentration fields on transformation hysteresis and mechanical stability; produced parameter guidelines for collaborating experimentalists.
- Implemented GPU-accelerated phase-field code in C with OpenACC on the Ohio Supercomputer Center; managed large parametric job sweeps via SLURM for composition and load studies.

Earlier project (co-author): Crystallite-Size Effects on Phase Transformations in WO_3

- Synthesized WO_3 nanocrystals via sol-gel method; quantified crystallite size distributions from TEM micrographs to correlate microstructure with low-temperature solid-solid phase transformation behavior.

Team Member, "ML for Superconductivity Prediction", The Ohio State University *Spring 2026*

MATSCEN 7193.02 (Machine Learning for Materials Research), team project (4 members)

- Built XGBoost and GPR pipeline to predict superconducting parameters (λ , T_c) from Eliashberg spectral functions $\alpha^2 F(\omega)$ on the Gibson *et al.* (*npj Comput Mater*, 2026) dataset of 806 DFT-validated superconductors; designed physics-informed features (frequency-weighted $\alpha^2 F/\omega$ bins) aligned with the Eliashberg integral.
- Tuned GPR with Matérn ($\nu = 1.5, 2.5$) + White kernels to capture spectral discontinuities; benchmarked spectral bin resolution (50, 200, 800) against generalization to characterize the curse-of-dimensionality trade-off, identifying physics-informed binning as a more robust route than naive high-resolution sampling.

Teaching Experience

Graduate Teaching Assistant, The Ohio State University

Oct 2023 – Oct 2024

- Instructed two undergraduate courses: *Introduction to Engineering Materials* and *Modeling and Simulation Lab II*.
- Led weekly recitations, held office hours, and graded assignments and exams for approximately 40 students per course.

Conference Presentations

- **TMS 2027** (Mar 2027). Oral (accepted): *Dilute Compositional Modulation as a Design Lever for Martensitic Transformation in NiTi Shape Memory Alloys*. **Chen Z.**, Sriram H., Feng L., Yu T., Zhu J., Rajagopalan J., Wang Y.
- **TMS 2027** (Mar 2027). Oral (accepted): *Coupling Grain Structure and Compositional Gradients to Control Variant Selection in Polycrystalline NiTi*. **Chen Z.**, Wang Y.
- **TMS 2027** (Mar 2027). Oral (accepted): *AI-Enabled Inverse Design of Superelastic Materials through Phase-Field Simulations and Machine Learning*. **Chen Z.**, Hu K., Tan R., Zhang T., Wang Y.
- **TMS 2026** (Mar 2026). Poster: *Tailoring Superelastic Behavior in NiTi SMAs via 1D Compositional Modulation: A Calibrated Phase-Field Framework*. **Chen Z.**, Wang Y.
- **MS&T 2025** (Oct 2025). Oral: *Tailoring Mechanical Properties of NiTi Shape Memory Alloys Through Key Parameters in One-Dimensional Compositional Modulation*. **Chen Z.**, Wang Y.
- **TMS 2025** (Mar 2025). Poster: *Regulating Stress-Strain Behavior of NiTi Shape Memory Alloys via 1D Nanoscale Concentration Modulations*. **Chen Z.**, Sriram H., Feng L., Wang Y.
- **MS&T 2023** (Oct 2023). Oral: *Fine-tuning Superelastic Behavior of NiTi SMAs via Nanoscale Concentration Modulation Created by Ni_4Ti_3 Nanoprecipitate Dissolution*. **Chen Z.**, Sriram H., Feng L., Wang Y.

Professional Service

Reviewer, *Shape Memory and Superelasticity* (ASM International / Springer)

2026 – Present

Reviewer, *Metallurgical and Materials Transactions A* (TMS / ASM International / Springer) *2026 – Present*

Technical Skills

Programming: Python (scikit-learn, PyTorch), C, MATLAB, Fortran, Bash

HPC & GPU: GPU acceleration (CUDA, OpenACC), SLURM, Linux, Git, Ohio Supercomputer Center

Simulation: Phase Field Modeling, Shear Transformation Zone Kinetic Monte Carlo, CALPHAD (Thermo-Calc), Molecular Dynamics, Constitutive Modeling

Machine Learning: Gaussian Process Regression, Bayesian optimization, XGBoost, LASSO regression, Convolutional Neural Networks, Generative Models (Conditional Variational Autoencoders, VAE-GANs), Physics-Informed Neural Networks (PINNs)

Data & Visualization: OVITO, ParaView, PyVista, JupyterLab, MATLAB Advanced Plotting

Domain Expertise: Shape memory alloys, martensitic transformation, ICME, materials informatics

Languages: English (Fluent), Mandarin (Native)

References available upon request.